

H.M. SARWER ALAM

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PROFESSIONAL SUMMARY

Software Engineer with experience in Python, Django, FastAPI, C# .NET, SQL, and machine learning. Focused on software engineering, ML/AI systems, backend development, medical image analysis, explainable AI, RAG systems, and applied AI development.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL, C, C#, HTML, CSS, Django, FastAPI

ML/AI: TensorFlow, PyTorch, scikit-learn, OpenCV, Random Forest, XGBoost, SVM, Logistic Regression, KNN, Decision Trees

Tools: GitHub, Ubuntu, CUDA, Docker Basic, Microsoft Office

Core: ML Model Development and Deployment, Problem Solving, Teamwork, Communication, Adaptability

EXPERIENCE

BNX Network Inc.

April 2025 – Present

Software Engineer Junior, formerly Trainee Software Engineer

Naperville, IL, USA; Dhaka, Bangladesh

- Contributed to software projects including [SecVA Protector](#), [iConnectNow](#), IT Bot, and MSP-related systems under the guidance of senior engineers.
- Developed ML/AI solutions, including model training using Random Forest, XGBoost, and SVM.
- Built backend services using FastAPI and custom TCP-based server-client architecture.
- Built web applications using Python Django, HTML, and CSS; collaborated on C# .NET development, testing, validation, and debugging.

BRAC University

May 2023 – April 2024

Student Tutor, Undergraduate Teaching Assistant, CSE

Dhaka, Bangladesh

- Provided weekly tutoring on programming, debugging, and problem-solving; evaluated lab assignments with feedback.

PROJECTS

CiteWeave-DuckSummarizer

Personal Project

Python, BeautifulSoup, SentenceTransformer, TF-IDF, DuckDuckGo Search

[GitHub](#)

- Built a web summarization tool using hybrid summarization, embeddings, TF-IDF, and cosine similarity filtering.

Ask My PDF – Local RAG Chatbot

Personal Project

Python, FAISS, Sentence Transformers, PyTorch, Transformers, TinyLlama

[GitHub](#)

- Built an offline PDF chatbot using FAISS retrieval, local LLM inference, embedding, chunking, and RAG pipeline.

Predicting Heart Disease Using Machine Learning

University Project

Python, scikit-learn, Pandas, Matplotlib, Logistic Regression, Decision Trees, KNN, SVM

[GitHub](#)

- Implemented ML models and performed data preprocessing using Pandas and NumPy.

Random Image Classification

University Group Project

Python, ResNet50, AlexNet, Inception Net, Custom Model

[GitHub](#)

RESEARCH AND PUBLICATIONS

- 2 published journal papers** in IEEE JBHI and Emerging Science Journal on medical image analysis, explainable AI, and deep learning.
- 1 submitted conference paper** on semi-supervised federated learning using pseudo-labeling.

EDUCATION

BRAC University

2020 – 2024

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

- CGPA: 3.88/4.00; Vice Chancellor's List for 4 semesters; Dean's List for 5 semesters.

Bir Shreshtha Noor Mohammad Public College

Dhaka, Bangladesh

HSC: GPA 5.00/5.00, 2019 | SSC: GPA 5.00/5.00, 2017

AWARDS AND LEADERSHIP

- Best Thesis Award – Spring 2024**, BRAC University, February 28, 2025.
- Senior Executive – Communications and Marketing**, BRAC University Computer Club, August 2021 – August 2023.
- House Vice Captain – MJ House**, Bir Shreshtha Noor Mohammad Public College, January 2018 – April 2019.